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Generative AI - Assignment 1

A Comparative Analysis of Generative Models for Image
Synthesis and Anomaly Detection

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Abstract

We performed a comprehensive implementation and comparative analysis of three distinct generative models: Generative Adversarial Networks (GANs), Conditional GANs (CGANs), and Variational Autoencoders (VAEs). Each model was trained and evaluated for image synthesis on two benchmark datasets, MNIST and CIFAR-10, to assess performance across different levels of data complexity. The generative quality was quantitatively measured using Fréchet Inception Distance (FID) and Kernel Inception Distance (KID), which highlighted the superior image fidelity of the standard GAN architecture. Furthermore, the VAE's utility for unsupervised anomaly detection was demonstrated on a real-world credit card fraud dataset. A detailed evaluation, supported by a confusion matrix and yielding a balanced F1-score of 0.93, confirmed its efficacy in identifying rare fraudulent transactions. This study provides a data-driven comparison, elucidating the specific strengths and weaknesses of each architecture for their respective tasks.

1 Introduction

We explore three foundational generative architectures: GANs, CGANs, and VAEs. The primary objective is to conduct a rigorous, empirical comparison by training each model on both the MNIST and CIFAR-10 datasets for image generation. This dual-dataset approach allows for an analysis of how each model's performance scales with data complexity. Performance is evaluated through two lenses: generative quality, measured quantitatively with FID and KID, and the practical application of VAEs for unsupervised anomaly detection on a credit card fraud dataset.

2 Methodology

2.1 Datasets and Visualization

Experiments were conducted on three datasets:

- **MNIST & CIFAR-10:** Standard image datasets used for the generative tasks. Pixel values were normalized to a range of $[-1, 1]$ to aid training stability.
- **Credit Card Fraud Dataset:** A highly imbalanced dataset containing transactional data, where the vast majority of instances are legitimate. This dataset was used exclusively for the anomaly detection task.

A visual preview of the image datasets is provided in [Figure 1](#).

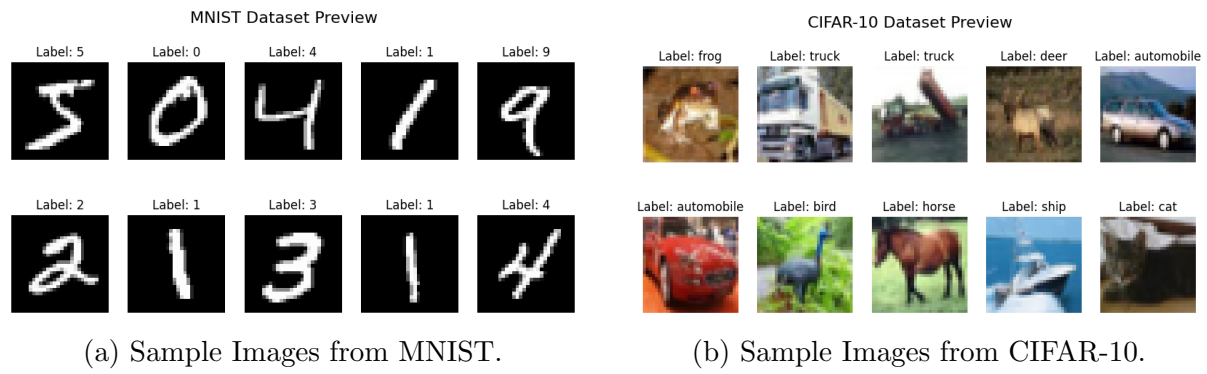


Figure 1: Image Dataset Preview.

2.2 Model Architectures

Deep convolutional neural networks formed the backbone of the image-based models (GAN, CGAN, VAE for image tasks). For the credit card fraud detection task, the VAE was composed of fully connected layers suitable for tabular data. The encoder maps a transaction's features to a latent space, and the decoder attempts to reconstruct them.

3 Experiments and Results

3.1 Part 1: Generative Adversarial Network (GAN)

The GAN model was trained on both image datasets. Its performance is characterized by the adversarial loss curves and the quality of the final generated images.

3.1.1 GAN on MNIST

On MNIST, the GAN's training was stable, generating sharp and diverse handwritten digits.

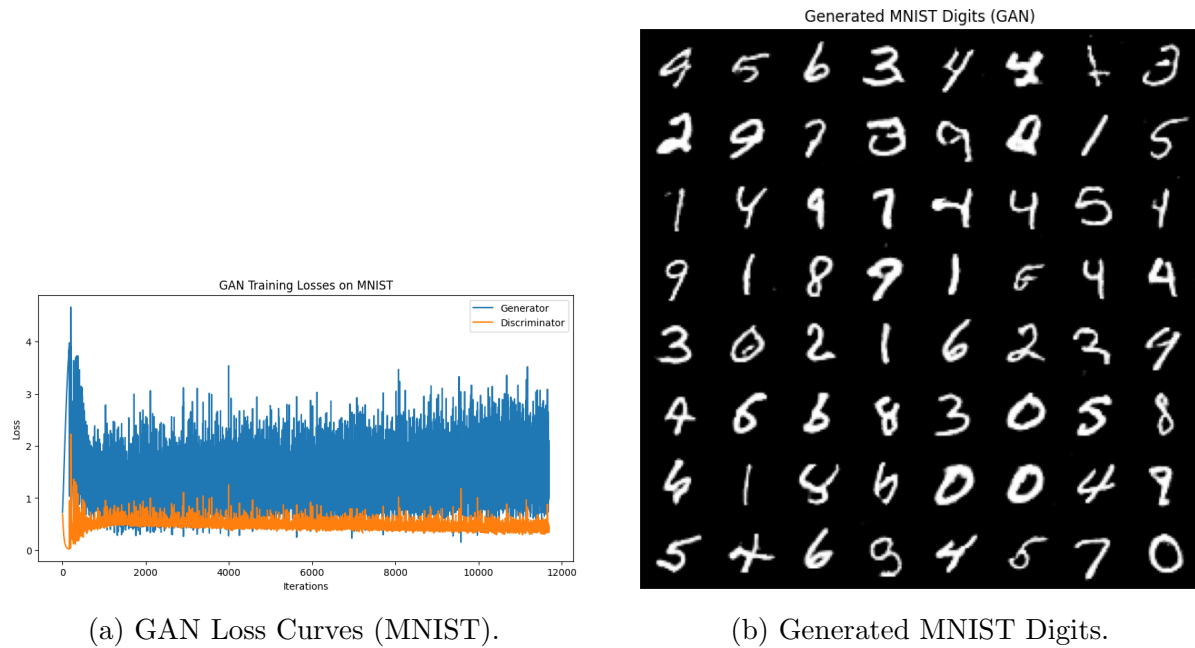


Figure 2: GAN Training and Generation Results on MNIST.

3.1.2 GAN on CIFAR-10

On the more complex CIFAR-10 dataset, the GAN training exhibited more oscillations in the loss curves. The generated images are more abstract and less coherent.

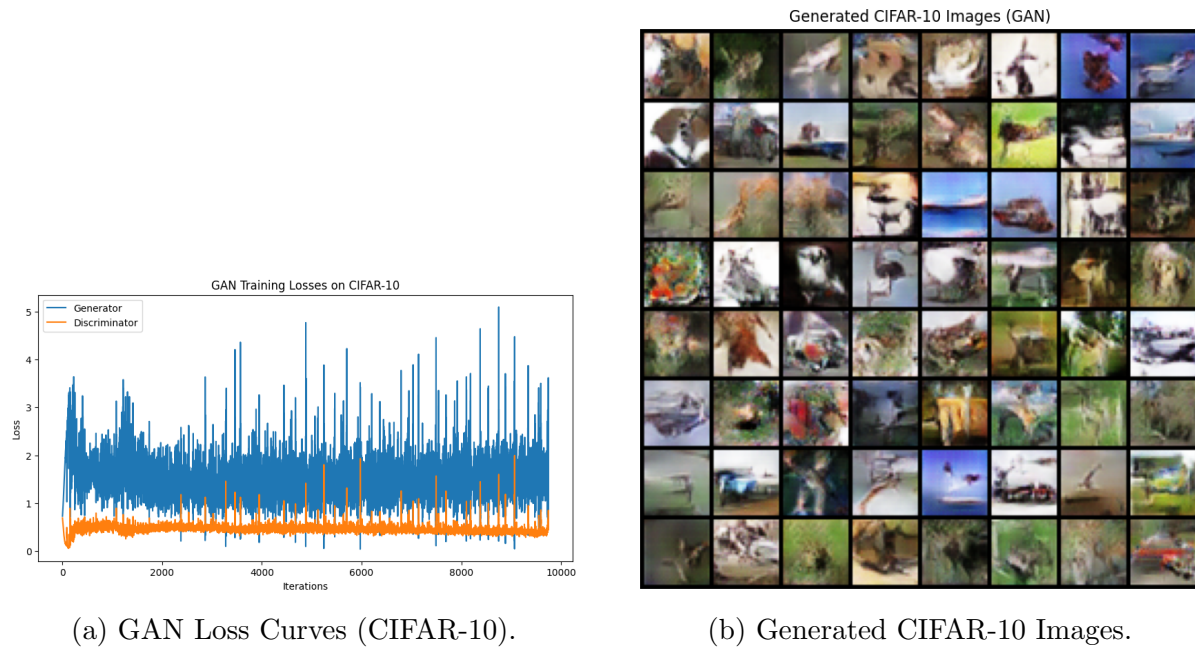


Figure 3: GAN Training and Generation Results on CIFAR-10.

3.1.3 Quantitative Evaluation of GAN

The FID and KID scores in Table 1 confirm the visual assessment. The scores for MNIST are significantly better (lower) than for CIFAR-10.

Table 1: Standard GAN Performance Metrics (Lower is better).

Dataset	FID	KID
MNIST	24.26	0.0123
CIFAR-10	70.14	0.0263

3.2 Part 2: Conditional GAN (CGAN)

The CGAN was trained to generate images conditioned on class labels.

3.2.1 CGAN on MNIST

The CGAN on MNIST successfully generated digits corresponding to the provided labels.

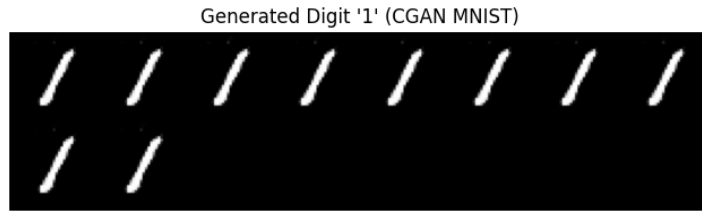


Figure 4: Generated MNIST Digits by Label using CGAN.

3.2.2 CGAN on CIFAR-10

On CIFAR-10, the CGAN struggled to generate distinct class-conditional images.

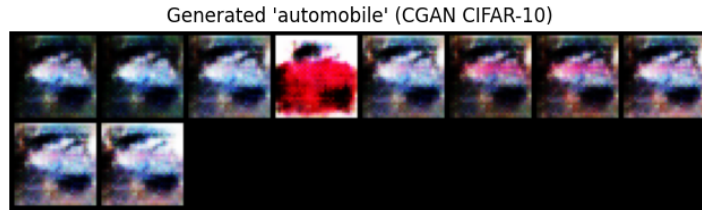


Figure 5: Generated CIFAR-10 Images by Label using CGAN.

3.2.3 Quantitative Evaluation of CGAN

The quantitative results for the CGAN, shown in Table 2, were unexpectedly high (worse) compared to the standard GAN, especially on MNIST. This suggests potential issues with training stability or mode collapse when introducing conditional information in this specific architecture.

Table 2: Conditional GAN Performance Metrics (Lower is better).

Dataset	FID	KID
MNIST	198.61	0.254
CIFAR-10	241.96	0.255

3.3 Part 3: Variational Autoencoder (VAE)

The VAE was evaluated on its ability to generate new image samples and learn a structured latent space.

3.3.1 VAE on MNIST

The VAE performed well on MNIST, generating plausible new digits and learning a well-clustered latent space.

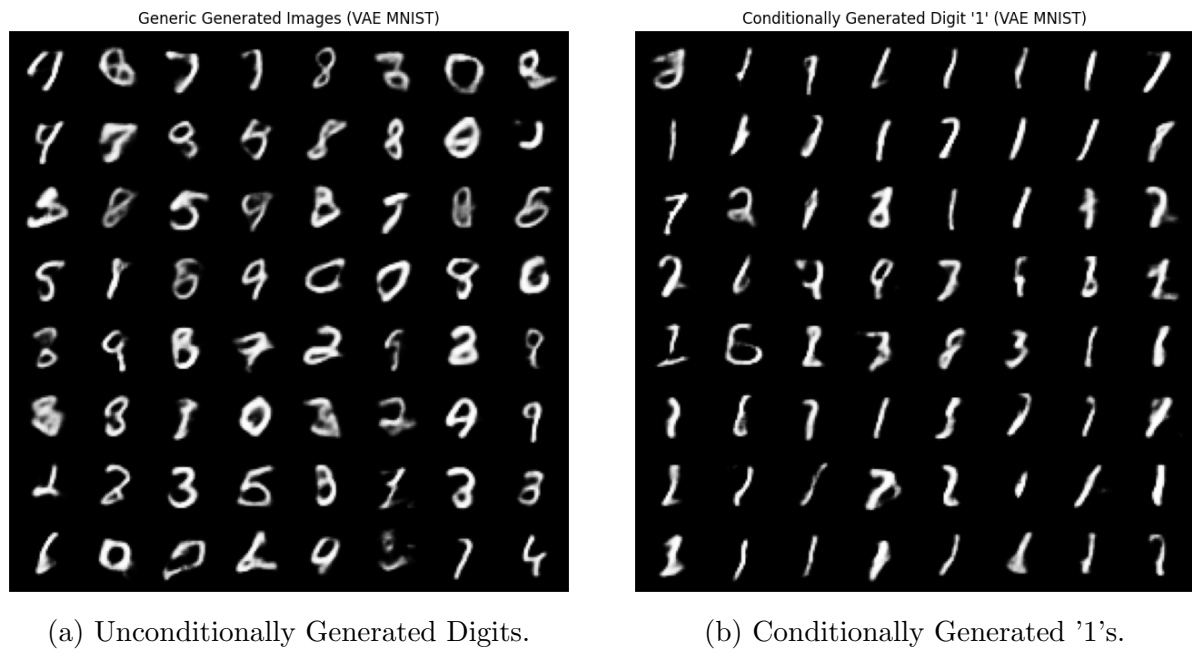


Figure 6: VAE Image Generation on MNIST.

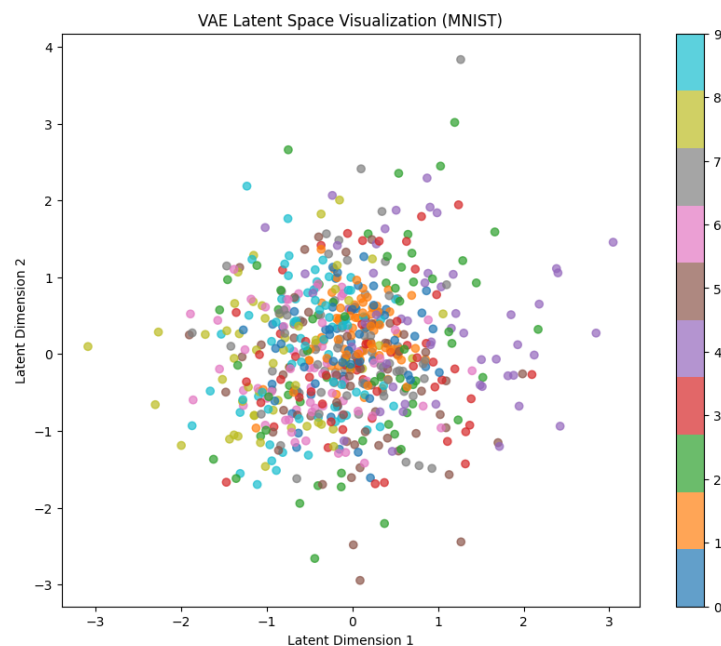


Figure 7: VAE Latent Space Visualization for MNIST.

3.3.2 VAE on CIFAR-10

On CIFAR-10, the VAE's performance degraded significantly. The generated images are very blurry and abstract, which can be attributed to several factors inherent to the VAE architecture and the dataset's complexity. The VAE's objective function, which includes a pixel-wise reconstruction loss (like Mean Squared Error), tends to average possibilities, leading to blurriness rather than sharp details. Furthermore, the KL divergence term regularizes the latent space, forcing it into a simple prior distribution (e.g., a Gaussian). This constraint is too restrictive for capturing the high variance of the CIFAR-10 dataset (color, objects, backgrounds), causing the model to sacrifice image fidelity. This struggle is quantitatively confirmed by the extremely high FID and KID scores. The latent space also shows much less distinct clustering compared to MNIST, reflecting the model's difficulty in learning a separable representation for the complex classes.

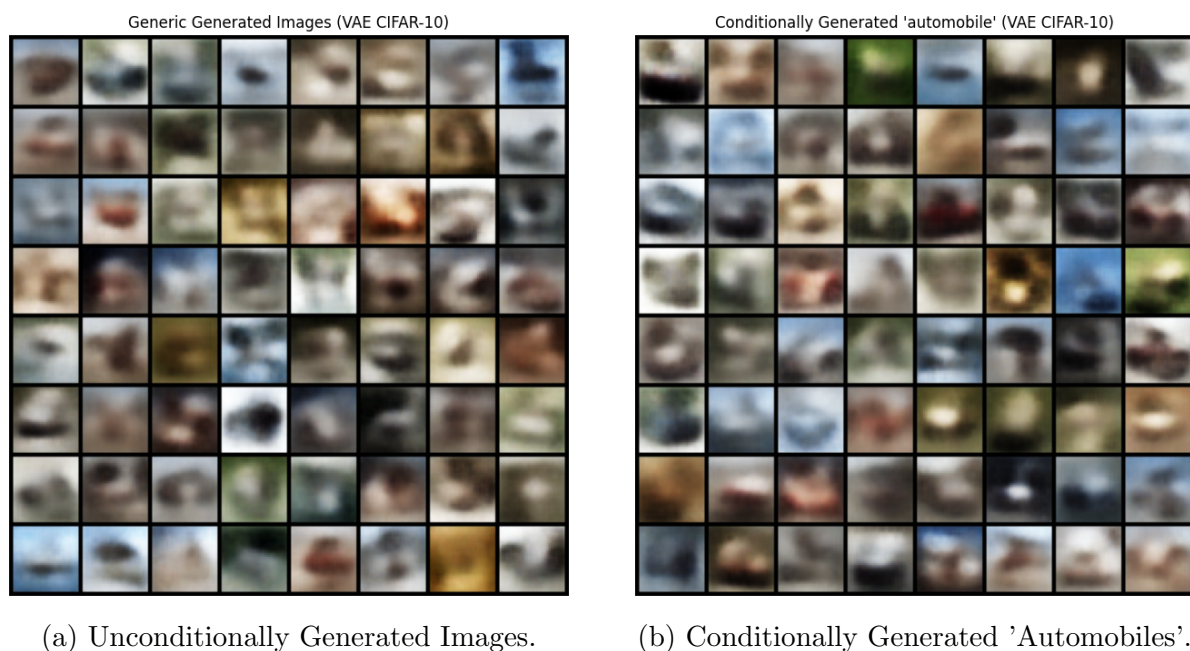


Figure 8: VAE Image Generation on CIFAR-10.

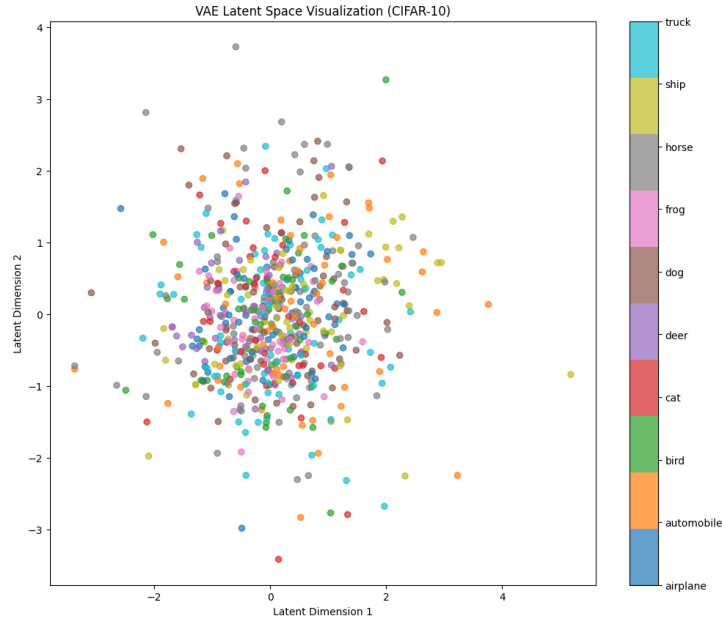


Figure 9: VAE Latent Space Visualization for CIFAR-10.

3.3.3 Quantitative Evaluation of VAE

The VAE’s generative quality, assessed using FID and KID in Table 3, was significantly worse than the GAN-based models, especially on CIFAR-10.

Table 3: Variational Autoencoder Performance Metrics (Lower is better).

Dataset	FID	KID
MNIST	88.35	0.0708
CIFAR-10	296.05	0.2716

3.4 Part 4: Comparative Evaluation and Anomaly Detection

3.4.1 Quantitative Comparison of All Models

Table 4 provides a comprehensive summary. The standard GAN emerges as the top-performing model for generative quality on both datasets. All models struggled significantly more with CIFAR-10 than with MNIST.

Table 4: Generative Model Performance Comparison (Lower is better).

Dataset	Model	FID	KID
MNIST	GAN	24.26	0.0123
	CGAN	198.61	0.254
	VAE	88.35	0.0708
CIFAR-10	GAN	70.14	0.0263
	CGAN	241.96	0.255
	VAE	296.05	0.2716

3.4.2 VAE for Anomaly Detection on Credit Card Fraud Data

The VAE was trained on the non-fraudulent transactions from the credit card dataset. The goal was to identify fraudulent transactions, which are rare, by flagging them as anomalies with high reconstruction error. The error distribution in Figure 10 shows a clear separation between the errors for legitimate and fraudulent transactions. The confusion matrix in Figure 11 visually represents the classification performance.

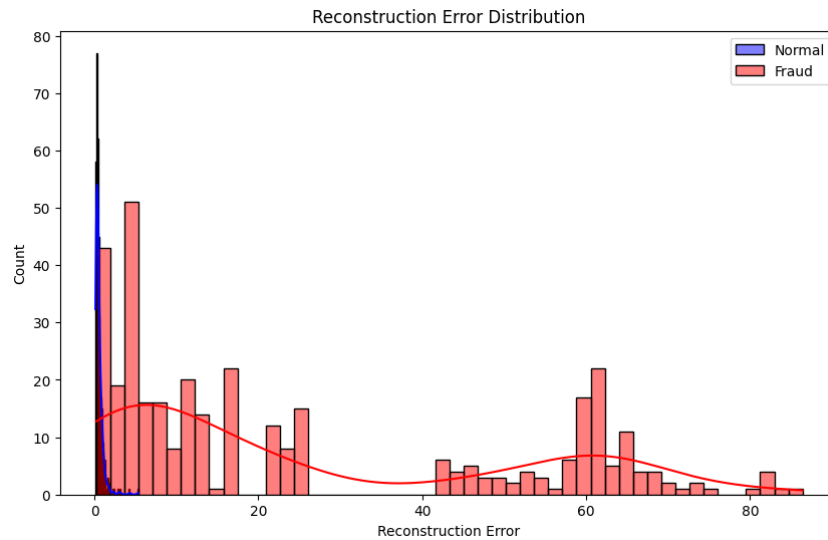


Figure 10: Reconstruction Error Distribution for Legitimate vs. Fraudulent Transactions.

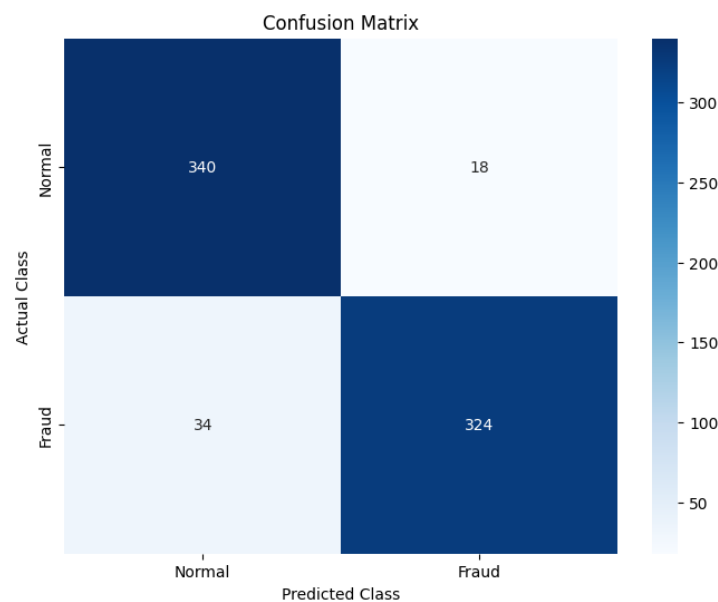


Figure 11: Confusion Matrix for VAE Anomaly Detector on Credit Card Fraud Data.

The final classification metrics in Table 5 quantify the model's strong and balanced performance, achieving an overall accuracy of 93%.

Table 5: Anomaly Detection Performance Metrics on Credit Card Fraud Data using VAE.

Class	Precision	Recall	F1-Score
Normal	0.91	0.95	0.93
Fraud	0.95	0.91	0.93
Accuracy	0.93		

4 Discussion and Conclusion

We successfully demonstrated the implementation and rigorous evaluation of GANs, CGANs, and VAEs. The quantitative FID/KID metrics were crucial for objectively comparing generative capabilities. The results decisively showed that the standard GAN architecture produced the highest-fidelity images on both datasets. The CGAN, while effective qualitatively on MNIST, performed poorly in the quantitative metrics, suggesting that the conditioning mechanism may have introduced instability.

The VAE, despite being the weakest in image generation quality, proved to be a robust and quantitatively superior framework for unsupervised anomaly detection. When applied to the credit card fraud dataset, it achieved a strong accuracy of 93% and a balanced F1-score of 0.93 for both normal and fraudulent classes. The excellent balance between precision (0.95) and recall (0.91) for the fraudulent class is particularly noteworthy, as it indicates the model is effective at both identifying fraud and minimizing false alarms. These findings confirm that the choice of generative model is highly task-dependent.